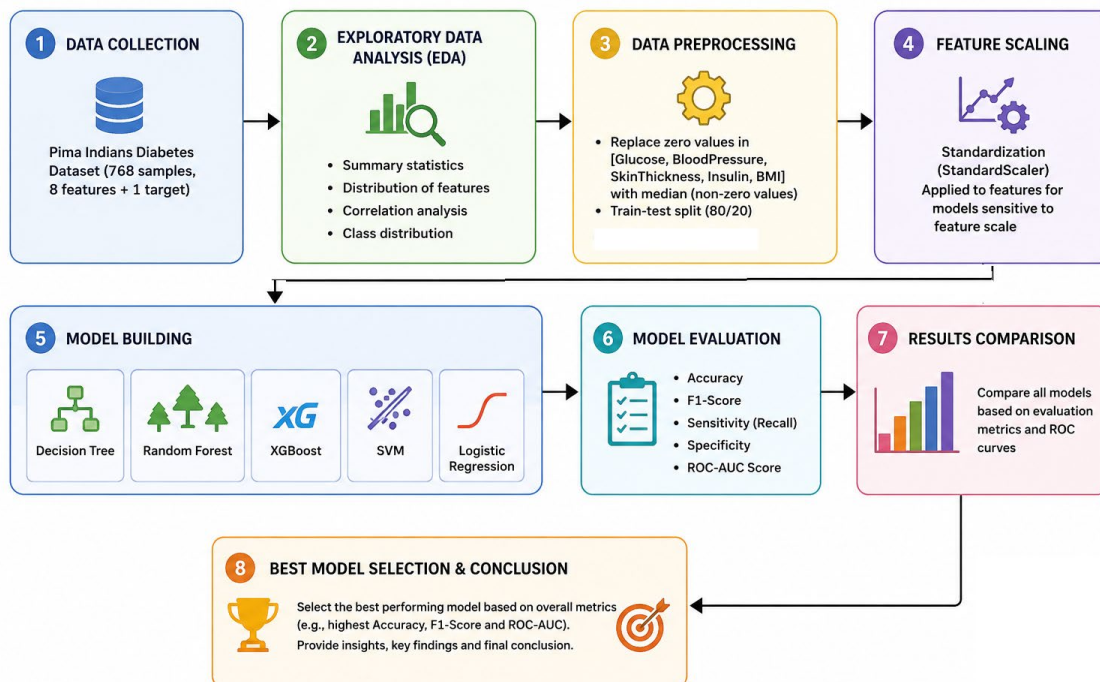


Objective

Develop and compare machine learning models for predicting diabetes using the Pima Indians Diabetes dataset.

DIABETES PREDICTION PROJECT – BLOCK DIAGRAM



Dataset

768 samples, 8 numerical features and one binary target (Outcome).

Descriptive statistics

Descriptive statistics are used to summarize and describe variables for the data. It includes mean, std, min, max, median, 25%, 50%, 75%.

Preprocessing

Zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI were replaced with the median of non-zero values. Data were split into 80% training and 20% testing (stratify=Yes). Feature scaling (Normalization) was applied where appropriate.

Models

Logistic Regression, Decision Tree, Random Forest, SVM and XGBoost (5 models).

Evaluation

Accuracy, F1-Score, Sensitivity, Specificity, ROC-AUC, confusion matrices and ROC curves were used.

Analysis

Decision Tree achieved the highest Accuracy (78.57%), F1-Score (0.6916) and Sensitivity (68.52%). XGBoost achieved the highest ROC-AUC (0.8246). Random Forest also performed well, while Logistic Regression showed the weakest performance. Tree-based models captured nonlinear relationships more effectively than the linear model.

Model	Accuracy	F1	Sensitivity	Specificity	ROC-AUC
Decision Tree	0.7857	0.6916	0.6852	0.8400	0.7887
XGBoost	0.7727	0.6392	0.5741	0.8800	0.8246
Random Forest	0.7597	0.6263	0.5741	0.8600	0.8142
SVM	0.7403	0.6000	0.5556	0.8400	0.7964
Logistic Regression	0.7078	0.5455	0.5000	0.8200	0.8130

Feature Importance Analysis

In order to fine the effect of each feature on the output prediction and to find more effective features, importance analysis has been done using random forest model. The result presents that 'Glucose' has the most effect on the output.

Conclusion

All models exceeded the required 70% accuracy. Decision Tree provided the best balance between overall accuracy and detection of diabetic patients and is recommended as the final model.

Considering the balance between overall accuracy and the ability to correctly detect diabetic patients, the **Decision Tree** can be considered the most appropriate model for this dataset. However, if maximizing discrimination across thresholds is the primary objective, **XGBoost** would be a strong alternative due to its superior ROC-AUC.